LN4 Brownian Motion

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Spring 2023

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Textbook: Stochastic Calculus for Finance II: Continuous-Time Models: Chapter 3

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Symmetric Random Walk

Consider an infinite sequence of coin tosses $\omega = \omega_1\omega_2\omega_3\cdots$.

$$X_j = \begin{cases} 1 & if \quad \omega_j = H, w.p \quad p \\ -1 & if \quad \omega_j = T, w.p \quad q \end{cases} \tag{1}$$

Let $M_0 = 0, M_k = \sum_{j=1}^{k} X_j, k = 1, 2, \cdots$. If $p = q = \frac{1}{2}$, M_k is a symmetric random walk.

Increments of the Symmetric Random Walk

If we choose non-negative integers $0 = k_0 < k_1 < k_2 < \cdots < k_m$, the random variables

$$M_{k_{i+1}} - M_{k_i} = \sum_{j=k_i+1}^{k_{i+1}} X_j$$

are independent increments.

$$Var(M_{k_{i+1}} - M_{k_i}) = \sum_{j=k_i+1}^{k_{i+1}} Var(X_j) = \sum_{j=k_i+1}^{k_{i+1}} 1 = k_{i+1} - k_i.$$

The variance of the symmetric random walk accumulates at rate one per unit time.

Martingale property of the Symmetric Random Walk

We choose non-negative integers $k < l$ and compute

$$\mathbb{E}(M_l|\mathcal{F}_k) = \mathbb{E}(M_l - M_k + M_k|\mathcal{F}_k) \quad = \mathbb{E}(M_l - M_k|\mathcal{F}_k) + M_k = M_k \tag{2}$$

Quadratic Variation of the Symmetric Random Walk

The quadratic variation up to time k is defined to be

$$[M, M]_k = \sum_{j=1}^{k} (M_j - M_{j-1})^2 = k$$

We note that $[M, M]_k$ is the same as $Var(M_k)$, but the computations of these two quantities are different. $Var(M_k)$ is computed by taking an average over all paths, taking their probability into account. By contrast, $[M, M]_k$ is computed path by path.

Scaled Symmetric Random Walk

Define the scaled symmetric random walk

$$W^{(n)}(t) = \frac{1}{\sqrt{n}} M_{nt},$$

provided nt is an integer.

Like the random walk, the scaled random walk has independent increments. Furthermore, if $0 \leq s \leq t$ are such that ns and nt are integers, then

$$\mathbb{E}(W^{(n)}(t) - W^{(n)}(s)) = 0, \ \ Var(W^{(n)}(t) - W^{(n)}(s)) = t - s$$

Scaled Symmetric Random Walk

The martingale property:

$$\mathbb{E}(W^{(n)}(t)|\mathcal{F}(s)) = \mathbb{E}(W^{(n)}(t) - W^{(n)}(s) + W^{(n)}(s)|\mathcal{F}(s)) = W^{(n)}(s)$$

Finally, we consider the quadratic variation.

$$\begin{aligned}[W^{(n)}, W^{(n)}](t) &= \sum_{j=1}^{nt} \left[W^{(n)}(\frac{j}{n}) - W^{(n)}(\frac{j-1}{n}) \right]^2 \\ &= \sum_{j=1}^{nt} \left[\frac{1}{\sqrt{n}} X_j \right]^2 = \sum_{j=1}^{nt} \frac{1}{n} = t \end{aligned} \tag{3}$$

Limiting Distribution of the Scaled Random Walk

Theorem 1

Fix $t \geq 0$. As $n \to \infty$, the distribution of the scaled random walk $W^{(n)}(t)$ evaluated at time t converges to the normal distribution with mean zero and variance t.

Binomial asset-pricing model $\to$ BS model.

Brownian Motion

Definition 1

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. For each $\omega \in \Omega$, suppose there is a continuous function $W(t)$ of $t \geq 0$ that satisfies $W(0) = 0$ and that depends on ω. Then $W(t), t \geq 0$, is a Brownian motion if for all $0 = t_0 < t_1 < \cdots < t_m$ the increments

$$W(t_1), W(t_2) - W(t_1), \cdots, W(t_m) - W(t_{m-1})$$

are independent and each of these increments is normally distributed with

$$\mathbb{E}(W(t_{i+1}) - W(t_i) = 0, \ \ Var(W(t_{i+1}) - W(t_i) = t_{i+1} - t_i.$$

Filtration for Brownian Motion

Definition 2

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space on which is difined a BM $W(t), t \geq 0$. A filtration for the Brownian motion is a collection of σ-algebras $\mathcal{F}(t)$, $t \geq 0$, satisfying:

(1) *(Information accumulates) For $0 \leq s < t$, every set in $\mathcal{F}(s)$ is also in $\mathcal{F}(t)$. In other words, there is at least as much information available at the later time $\mathcal{F}(t)$ as there is at the earlier time $\mathcal{F}(s)$.*

(2) *(Adaptivity) For each $t \geq 0$, the BM $W(t)$ at time t is $\mathcal{F}(t)$-measurable. In other words, the information available at time t is sufficient to evaluate the Brownian motion $W(t)$ at that time.*

(3) *(Independence of future increments) For $0 \leq t < u$, the increment $W(u) - W(t)$ is independent of $\mathcal{F}(t)$. In other words, any increment of the BM after time t is independent of the information available at time t.*

Brownian motion

Theorem 2

Brownian motion is a martingale.

Proof: Let $0 \leq s \leq t$ be given. Then

$$\mathbb{E}(W(t)|\mathcal{F}(s)) = \mathbb{E}(W(t) - W(s) + W(s)|\mathcal{F}(s)) = W(s).$$

Quadratic Variation

For BM, there is no natural step size. If we are given $T > 0$, we could choose a step size, say $\frac{T}{n}$ for some large n, and compute the QV by

$$\sum_{j=0}^{n-1} \left[W\left(\frac{(j+1)T}{n}\right) - W\left(\frac{jT}{n}\right) \right]^2.$$

We are interested in this quantity as $n \to \infty$.

First-Order Variation

We first choose a partition $\Pi = \{t_0, t_1, \cdots, t_n\}$ of $[0, T]$ and $0 = t_0 < t_1 < \cdots < t_n = T$. The maximum step size is denoted by $\|\Pi\| = \max_{j=0,\cdots,n-1}(t_{j+1} - t_j)$. We then define

$$FV_T(f) = \lim_{\|\Pi\| \to 0} \sum_{j=0}^{n-1} |f(t_{j+1}) - f(t_j))|.$$

Quadratic Variation

Definition 3

Let $f(t)$ be a function defined for $0 \leq t \leq T$. The quadratic variation of f up to time T is

$$[f, f](T) = \lim_{\|\Pi\| \to 0} \sum_{j=0}^{n-1} [f(t_{j+1}) - f(t_j)]^2,$$

where $\Pi = \{t_0, t_1, \cdots, t_n\}$ and $0 = t_0 < t_1 < \cdots < t_n = T$.

Quadratic Variation

Remark Suppose f has a continuous derivative. Then

$$\sum_{j=0}^{n-1}[f(t_{j+1}) - f(t_j)]^2 = \sum_{j=0}^{n-1} |f'(t_j^*)|^2 (t_{j+1} - t_j)^2$$

$$\leq \|\Pi\| \cdot \sum_{j=0}^{n-1} |f'(t_j^*)|^2 (t_{j+1} - t_j),$$

Thus,

$$[f, f](T) \leq \lim_{\|\Pi\| \to 0} \left[\|\Pi\| \cdot \sum_{j=0}^{n-1} |f'(t_j^*)|^2 (t_{j+1} - t_j) \right]$$

$$= \lim_{\|\Pi\| \to 0} \|\Pi\| \cdot \int_0^T |f'(t)|^2 dt = 0$$

So we don't use it often in ordinary calculus.

Quadratic Variation of A Brownian Motion

Theorem 3

Let W be a Brownian motion. Then $[W, W](T) = T$ for all $T \geq 0$ almost surely.

Proof Let $\Pi = \{t_0, t_1, \cdots, t_n\}$ be a partition of $[0, T]$. Define the sampled quadratic variation to be

$$Q_\Pi = \sum_{j=0}^{n-1} (W(t_{j+1}) - W(t_j))^2.$$

Then

$$\begin{aligned} \mathbb{E}(Q_\Pi) &= \sum_{j=0}^{n-1} \mathbb{E}(W(t_{j+1}) - W(t_j))^2 = \sum_{j=0}^{n-1} Var(W(t_{j+1}) - W(t_j)) \\ &= \sum_{j=0}^{n-1} (t_{j+1} - t_j) = T. \end{aligned} \tag{4}$$

Proof of Theorem 3 Continued

Moreover,

$$
\begin{aligned}
&Var\left[(W(t_{j+1}) - W(t_j))^2\right] \\
&= \mathbb{E}\left[\left((W(t_{j+1}) - W(t_j))^2 - (t_{j+1} - t_j)\right)^2\right] \\
&= \mathbb{E}\left(W(t_{j+1}) - W(t_j)\right)^4 - 2(t_{j+1} - t_j)\mathbb{E}\left(W(t_{j+1}) - W(t_j)\right)^2 + (t_{j+1} - t_j)^2 \\
&= 3(t_{j+1} - t_j)^2 - 2(t_{j+1} - t_j)^2 + (t_{j+1} - t_j)^2 \\
&= 2(t_{j+1} - t_j)^2.
\end{aligned} \tag{5}
$$

and

$$
Var(Q_\Pi) = \sum_{j=0}^{n-1} 2(t_{j+1} - t_j)^2 \leq 2\|\Pi\| T \to 0.
$$

Quadratic Variation

Remark

$$
\begin{aligned}
(W(t_{j+1}) - W(t_j))^2 &\approx t_{j+1} - t_j \\
dW(t)dW(t) &= dt \\
dW(t)dt &= 0 \\
dtdt &= 0
\end{aligned}
\tag{6}
$$

Markov Property

Theorem 4

Let $W(t), t \geq 0$ be a Brownian motion and let $\mathcal{F}(t), t \geq 0$, be a filtration for this BM. Then $W(t)$ is a Markov process.